

Saaras Pakanati

Cincinnati, Ohio, USA

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Education

Rice University, PhD Mechanical Engineering (incoming student) Advisor: Prof. Daniel Preston	August 2026 – Present GPA –/4.00
University of Cincinnati, BS Mechanical Engineering, <i>Summa Cum Laude</i> <i>Minor in Mathematics</i> Advisor: Prof. Kishan Bellur	August 2021 – May 2026 GPA 3.96/4.00

Research Experience

Undergraduate Research Assistant UC Lab for Interfacial Dynamics, advised by Dr. Kishan Bellur Dept. of Mechanical & Materials Engineering, University of Cincinnati	August 2024 – May 2026
• Numerical & Experimental study of multiphase interfaces and associated phenomenon. • Modeled multiscale cryogenic fuel evaporation using custom UDFs in ANSYS Fluent. • Designed & built an interferometry setup to study instabilities in liquid thin films. • Assisting in the study of the effect of temperature-sensitive surfactants on interfacial dynamics.	

Peer-Reviewed Publications

[†] denotes equal contribution

2. A. Sarchami, **S. Pakanati**, A. Yasin, M. Jog, K. Bellur, *An experimental study of interfacial dynamics control using temperature-sensitive surfactants*, Langmuir, 2026
10.1021/acs.langmuir.5c06748
1. A. Yasin[†], **S. Pakanati**[†], K. Bellur, *A multiscale CFD model of evaporating Hydrogen menisci: Incorporating subgrid thin-film dynamics and in situ accommodation coefficients*, Fuels, 2026
10.3390/fuels7010003

Conference Presentations

* denotes presenting author

2. A. Sarchami, **S. Pakanati**, T. Enam, A. Yasin, K. Bellur*, *Stability and Contact Line Dynamics of Evaporating Thin Liquid Films*, 11th ASTFE Thermal and Fluids Engineering Conference, Tempe, Arizona, March 9–12, 2026
1. A. Sarchami*, **S. Pakanati**, K. Bellur, *Dynamic Optical Measurement of Evaporating Liquid Films*, ASME International Mechanical Engineering Congress & Exposition, Memphis, Tennessee, November 16–20, 2025 (Poster)

Honors, Awards, and Fellowships

Technical Excellence in Senior Design Project – 2nd place , Awarded by the College of Engineering and Applied Science	2026
Merchant, M. Eugene Scholarship , Awarded by the College of Engineering and Applied Science	2026
Armstrong Discovery Fellowship (2×) , Awarded by the Armstrong Institute for Space, Technology, and Research	2025 & 2026
CEAS Annual Scholar – Undergraduate Research , Awarded by the College of Engineering and Applied Science	2025
Undergraduate Research co-op Fellowship (2×) , Awarded by the College of Engineering and Applied Science	2024 & 2025
UC Calculus Contest – 2nd place , Awarded by Cengage Publishing & UC Department of Mathematical Sciences	2022
CEAS International Outreach Scholarship , Awarded by the College of Engineering and Applied Science	2021
University of Cincinnati Global Scholarship , Awarded by the University of Cincinnati	2021

Teaching Experience

Grader Dept. of Mechanical & Materials Engineering, University of Cincinnati	August 2025 – December 2025
• Course(s): MECH 2010 (Thermodynamics) • Managed assignments and proctored exams for 150+ students.	

- Led in-class review sessions for Calculus I and II.
- Tutored students in 15 different Engineering, Mathematics, Physics, and Chemistry courses.

Industry Experience

New Product Development Mechanical Engineering co-op
MSA Safety, Cranberry Township, Pennsylvania

January 2024 – April 2024

- Executed Finite Element Analysis in Ansys to validate design decisions for safety devices.
- Optimized prototype testing of Technology Readiness Level (TRL) 1 gas detection technologies by developing and applying Design of Experiments (DOE) methods using Minitab.
- Collaborated with Electrical and Software teams for cross-functional design.

Engineering co-op
Richards Industrials, Cincinnati, Ohio

August 2022 – December 2022 & May 2023 – August 2023

- Tested critical load-bearing components using destructive and non-destructive methods.
- Created technical drawings for shop floor fixtures using SolidWorks, collaborating with machinists.
- Developed tools and reports for engineering calculations, process optimization, and quality analysis.

Software Intern
Visual Collaboration Technologies, Inc., Bengaluru, Karnataka, India

June 2022 – July 2022

- Created a visualization tool for a CFD study using the dash framework in Python.
- Collaborated in an Agile development framework to code, following PEP8 standards.

Skills

Simulation: Ansys (Fluent, Static Structural, LS Dyna), Siemens simulation suite (Siemens NX, StarCCM+), OpenFOAM, ParaView, SolidWorks, LAMMPS

Programming: Python, MATLAB, C, C++, Bash, R Language, Minitab, LabView, $\text{\LaTeX}$

Experimental: Optical interferometry, image processing and analysis, flow control, semiconductor manufacturing, Design of Experiments

Certifications & Licenses

OASiS Foundations Program, *University of Cincinnati* – 2025 | [link to certificate](#)

MATLAB Fundamentals, *Mathworks* – 2023 | [link to certificate](#)

IBM Data Science Practitioner, *IBM* – 2022 | [link to certificate](#)

Six Sigma Yellow Belt, *The Council for Six Sigma Certification* – 2022 | Credential ID: h7L0bACNv5

NX – CAD Design Associate, *Siemens* – 2022 | Credential ID: 6763-164-884-2231

Extra-Curricular

President, Undergraduate Research Society
University of Cincinnati

September 2021 – May 2026

- President (2025–26), VP (2024–25) and Events Chair (2022–23 & 2023–24)
- Organized meetings with the executive team; managing presentations and logistics for regular outreach.
- Gave multiple presentations on undergraduate research at university classes, general meetings, and academic organizations.

Volunteer Tutor, Bearcat Buddies
Cincinnati, Ohio

October 2021 – December 2023

- Tutored students in the Cincinnati Public School system and assisted in the improvement of the content covered with the local in-charge.